

UC Name	UC_LMS_14 — Record New Book Acquisition
Summary	The system allows library staff to register newly acquired books along with their details and number of copies so that they become part of the digital catalog.
Dependency	UC_LMS_01 — Maintain Digital Catalog
Actors	Primary: Library Staff Secondary: Administrator
Preconditions	Staff member is authenticated Staff has permission to manage inventory
Description of the Main Sequence	• Step 1: Staff opens the acquisition module. • Step 2: Staff enters book details (title, author, ISBN, publisher, year). • Step 3: Staff specifies number of copies. • Step 4: System validates the input data. • Step 5: System saves the book and copies. • Step 6: System updates the catalog and makes the book visible.
Description of the Alternative Sequence	• Step 1: Staff enters invalid or incomplete data. Step 2: System displays validation errors. Step 3: Staff corrects and resubmits the data. • Step 1: System detects duplicate book entry. Step 2: System prompts confirmation to merge or cancel.
Non functional requirements	The system must validate and store data within 1–2 seconds. Data must not be lost during system interruptions. Only authorized staff can perform this action.
Postconditions	The new book and its copies are stored in the system and visible in the catalog.

UC Name	UC_LMS_15 — Track Copy Count per Title
Summary	The system allows staff to view and update the number of copies for each book title to maintain accurate inventory.
Dependency	UC_LMS_14
Actors	Primary: Library Staff
Preconditions	Book exists in the catalog . Staff is authorized
Description of the Main Sequence	• Step 1: Staff selects a book from the catalog. • Step 2: System displays current copy count. • Step 3: Staff updates the number of copies. • Step 4: System validates the update. • Step 5: System saves and updates total copies.
Description of the Alternative Sequence	• Step 1: Staff attempts to reduce copies below borrowed count. Step 2: System rejects the operation and displays an error. • Step 1: Invalid number entered. Step 2: System requests correction.
Non functional requirements	Updates must be processed in real time (<1 second). The system must ensure data consistency across all modules.
Postconditions	The system reflects the correct number of copies for the selected book.

UC Name	UC_LMS_16 — Manage Damaged or Lost Copies
Summary	The system allows staff to mark individual copies as damaged or lost to ensure accurate inventory and prevent their circulation.
Dependency	UC_LMS_15
Actors	Primary: Library Staff
Preconditions	Book copy exists, Staff has inventory permissions
Description of the Main Sequence	• Step 1: Staff selects a specific book copy. - Step 2: Staff marks the copy as damaged or lost. - Step 3: System updates the copy status. - Step 4: System removes the copy from available inventory.
Description of the Alternative Sequence	• Step 1: Staff selects a copy already marked inactive. - Step 2: System displays current status and prevents duplicate update. • Step 1: System fails to update due to database issue. - Step 2: System notifies staff and logs error.
Non functional requirements	Status updates must occur instantly and be logged for auditing. Only authorized users can modify copy status.
Postconditions	The selected copy is marked correctly and excluded from circulation.

UC Name	UC_LMS_17 — Automated Inventory Sync on Borrow/Return
Summary	The system automatically updates inventory and availability when a book is borrowed or returned to maintain real-time accuracy.
Dependency	UC_LMS_16
Actors	• Primary: System • Secondary: Student, Academic Staff , Library Staff
Preconditions	A valid borrow or return transaction occurs
Description of the Main Sequence	• Step 1: User performs borrow or return action. • Step 2: System processes the transaction. • Step 3: System updates copy status (Borrowed/Available). • Step 4: System updates available copy count. • Step 5: System reflects changes in the catalog.
Description of the Alternative Sequence	• Step 1: Transaction fails (e.g., book unavailable). Step 2: System cancels update and displays error. • Step 1: System encounters synchronization issue. Step 2: System retries or logs inconsistency.
Non functional requirements	Updates must occur within 1 second. The system must ensure data consistency and prevent race conditions.

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Postconditions	Inventory accurately reflects the result of the transaction.

UC Name	UC_LMS_18 — Track Status per Physical Copy
Summary	The system tracks and displays the status of each individual book copy to ensure precise inventory control.
Dependency	UC_LMS_15
Actors	• Primary: Library Staff • Secondary: System
Preconditions	Copies exist in the system
Description of the Main Sequence	• Step 1: Staff selects a book. • Step 2: System displays all copies with their statuses. • Step 3: System updates statuses automatically or through staff actions.
Description of the Alternative Sequence	• Step 1: System detects inconsistent status. • Step 2: System prompts correction or flags error.

Non functional requirements	Status updates must be synchronized across the system in real time. Data must remain consistent.
Postconditions	Each copy has an accurate and updated status.

UC Name	UC_LMS_19 — Formal Acquisition Request Process
Summary	The system allows users to submit requests for new books and supports an approval workflow for staff.
Dependency	UC_LMS_17
Actors	• Primary: Student, Academic Staff • Secondary: Library Staff, Administrator
Preconditions	User is logged in
Description of the Main Sequence	• Step 1: User submits a request with book details and justification. Step 2: System records the request. Step 3: Staff reviews the request. Step 4: Staff approves or rejects it. Step 5: System notifies the user.

Description of the Alternative Sequence	• Step 1: Request is incomplete. Step 2: System rejects submission and requests correction. • Step 1: Request is rejected. Step 2: System records rejection and notifies user. •
Non functional requirements	Requests must be processed within seconds. Notifications must be delivered reliably. Data must be securely stored.
Postconditions	The request is stored with its final decision.

UC Name	UC_LMS_20 — Generate Inventory Reports
Summary	The system allows administrators to generate reports on inventory status for decision-making purposes.
Dependency	
Actors	Primary: Administrator
Preconditions	User has reporting permissions
Description of the Main Sequence	• Step 1: User selects report type (inventory, damaged, borrowed). • Step 2: System retrieves relevant data.

	• Step 3: System generates the report. • Step 4: User views or downloads the report.
Description of the Alternative Sequence	• Step 1: No data available. Step 2: System displays empty report message. • Step 1: System fails to generate report. Step 2: System displays error and logs issue.
Non functional requirements	Reports must generate within 3 seconds. Data must be accurate and up-to-date. Access must be restricted to authorized users.
Postconditions	The user obtains a complete and accurate inventory report.

UC Name	UC_LMS_42 — Purchase New Books
Summary	Inventory Manager purchases new book copies from an external supplier, from selecting titles, issuing a Purchase Order, sending it to the supplier, receiving the shipment, verifying it, and closing the PO,so that the received copies can be cataloged and made available for loa

Dependency	UC_LMS_01 (auth), UC_LMS_19
Actors	Primary: Inventory Manger Secondary: Administrator
Preconditions	1. Inventory Manager is authenticated (UC_LMS_01).
Description of the Main Sequence	• Step 1: Inventory Manager selects “New Purchase Order”. • Step 2: Inventory Manager adds one or more PurchaseOrderLines (title, ISBN, quantity, unit price, supplier). • Step 3: System computes the PO total and validates it against the remaining procurement budget. • Step 4: Inventory Manager submits the PO for approval. PO status AwaitingApproval. • Step 5: Administrator reviews and approves the PO. PO status Approved.
Description of the Alternative Sequence	• A1 — Budget exceeded : System rejects submission and shows the remaining budget. Inventory Manager reduces quantities or cancels. PO returns to <i>Drafting</i>. • A2 — Administrator rejects PO : PO status → <i>Rejected</i>. Inventory Manager may edit and resubmit, or cancel.
Non functional requirements	Reports must generate within 3 seconds. Data must be accurate and up-to-date. Access must be restricted to authorized users.
Postconditions	The user obtains a complete and accurate inventory report.